

LexiLearn: Gamifying Learning for Dyslexia in a Novel Linguistic Context

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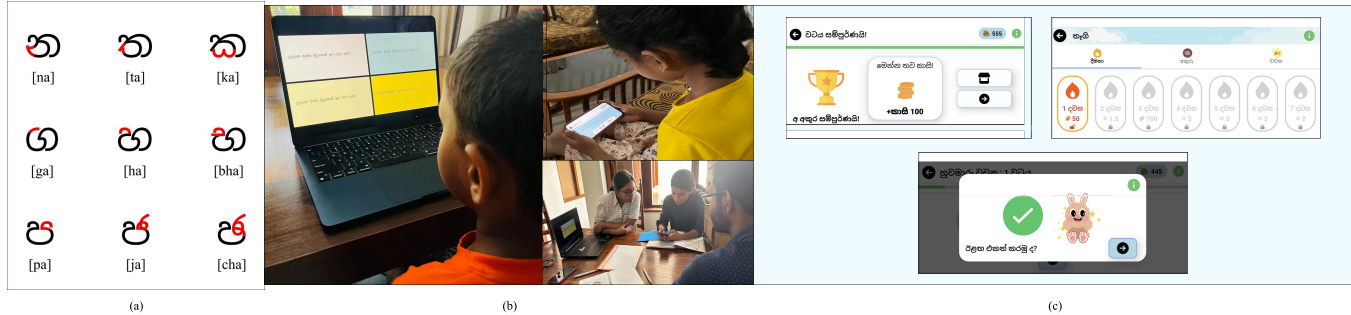


Figure 1: Figure (a) depicts the visual similarity of phonetically different characters in Sinhala, the primary spoken language in Sri Lanka, (b) illustrates students with dyslexia using LexiLearn app, as well as the co-design process with teachers, (c) represents the gamification elements – coins, daily streaks and avatars, integrated into the LexiLearn app.

Abstract

Gamification has shown significant potential to improve motivation and engagement in educational settings for students with learning disabilities like *dyslexia*. However, its application to linguistically and culturally unique contexts, such as Sinhala, which is a visually complex language spoken in Sri Lanka, remains underexplored. In this paper, we introduce LEXILEARN, a gamified mobile application designed to support Sinhala-speaking dyslexic learners. Developed through an iterative co-design process involving a dyslexia specialist, five teachers, eight students, and two language experts, LEXILEARN incorporates gamification elements such as coins, badges, avatars, levels, streaks, and continuous feedback across three core

activities. These activities target developing phonological awareness, word recognition, and reading comprehension for Sinhala language. A preliminary user study involving eight students with dyslexia revealed that LEXILEARN could potentially enhance their engagement in reading comprehension. Our work highlights how tailored educational tools can engage dyslexic learners and advance inclusive education in underrepresented linguistic settings.

CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods**; • **Applied computing** → **Education**.

Keywords

Dyslexia, Gamification, Co-Design, Inclusive Education

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1 Introduction

Dyslexia is a neurobiological learning disability that affects an individual's ability to read, write, and process information [27]. While it cannot be outgrown, early identification and intervention can help manage it effectively [3]. However, students with dyslexia¹ often face repeated academic failures, which can lead to self-doubt about their learning abilities, ultimately reducing their motivation to engage in academic tasks [24, 30].

Gamification, the use of game design elements in non-game contexts, has proven effective in increasing student motivation and engagement [10, 18, 25]. It enhances learning by leveraging *intrinsic motivators* like curiosity and challenges through elements such as leveling, animation, and engaging interfaces, as well as *extrinsic motivators* like rewards, praise, and competition that build confidence and encourage mastery. Several studies confirm its effectiveness in improving engagement and learning outcomes for students with dyslexia [5, 12, 13, 17, 33].

While gamification has shown significant promise in a global context, its application in diverse linguistic and cultural contexts remains underexplored. For visually complex languages like Sinhala² (one of the primary languages in Sri Lanka), tailored approaches are essential to address unique learning challenges. Sinhala, spoken by approximately 16 million people as a first language [11], is characterized by rounded shapes, distinct letter forms, and intricate vowel-consonant combinations [22, 39]. Unlike English, where letters may have multiple phonological sounds, Sinhala's complexity lies in its visual rather than phonological structure [22]. It consists of 60 characters featuring small and intricate differentiations (see Figure 1(a)), posing significant challenges for dyslexic learners [26, 34]. Resources specifically tailored to Sinhala-speaking students are scarce, and teachers often lack the training needed to address their specific needs [21]. Addressing these challenges requires localized, culturally relevant solutions that align with the language's structural characteristics.

Our work contributes to addresses these challenges in two ways. First, we introduce LEXILEARN, a gamified mobile application specifically designed for Sinhala-speaking students with dyslexia. By employing a co-design approach, involving discussions with dyslexia specialist teachers, students, and parents (shown in Figure 1(b)), we designed a system tailored to the needs of its target audience. LEXILEARN incorporates three core activities, Letter Mapping, Jumbled Words, and Reading Along, to target phonological awareness, word recognition, and reading comprehension. We integrate gamification elements such as coins, levels, badges, avatars, feedback and streaks (depicted in Figure 1(c)) to create an engaging and motivating learning experience.

Second, we present initial empirical findings on how gamification elements can be adapted to effectively motivate and engage students with dyslexia in a localized context. Our preliminary results show notable improvements in student engagement and positive perception of game elements such as coins, badges and avatars. By integrating gamification with localized content, our work highlights

the possibilities for supporting students with learning disabilities in underrepresented communities.

2 Background and Related Work on Gamification

Key Gamification Elements: Gamification elements have proven effective in enhancing engagement and learning outcomes in education. *Points* motivate learners by reinforcing behaviors and encouraging persistence, especially for those driven by external rewards [2]. *Levels* provide a structured learning path through scaffolding, starting with simpler challenges to build skills and confidence. Requiring mastery before progressing fosters achievement and skill development [2]. *Badges* serve as visual and textual incentives, highlighting accomplishments, setting clear goals, and boosting self-efficacy, making learning experiences more rewarding [19]. *Feedback* helps learners align their efforts with performance goals. When tailored and appropriately timed, it enhances motivation without overwhelming users, addressing their informational, emotional, and social needs [20]. Personalization elements like *Avatars* promote autonomy and self-expression, helping learners navigate digital spaces with confidence [2]. Finally, *Streaks* leverage principles like loss aversion to encourage consistent participation, fostering habits and sustained engagement [38].

Gamified Learning for Dyslexia: Previous research demonstrates that gamification effectively increases engagement among students with dyslexia [1, 5, 6, 8, 16, 29, 36]. Providing interactive and personalized learning experiences through elements like tailored content, have proven to be more effective than generic approaches, making Gamification particularly beneficial for students with dyslexia [13, 14]. For example, a twelve-week study using ClassDojo with seven students with dyslexia highlights the motivational impact of badges and the importance of personalized rewards to boost engagement [16]. Similarly, LexiPal, an Indonesian learning app, uses seven gamification elements such as, *levels, rewards, and feedback*. The initial study with 40 children with dyslexia showed improvements in their motor skills, emotion comprehension, and sentence production [5, 6, 36]. Another example, “Dyslexia Baca,” a Malay-language app, combines a multisensory approach with gamification elements like *levels, feedback, and incentives* to enhance alphabet recognition and engagement [9]. Currently, the only known Sinhala-language application for dyslexic learners, Arunalu [35] integrates basic gamified elements like levels and themes but lacks a strong reward system and effective incentives, limiting its ability to sustain engagement and foster achievement. These shortcomings highlight a critical gap in the availability of *effective, localized resources* for Sinhala-speaking students. Inspired by prior work, our research explores how incorporating gamification elements effectively motivates and engages students with dyslexia in a Sinhala language context.

3 Co-design and Implementation of LexiLearn

The co-design process, spanning 10 months, included a clinical psychologist, five teachers, six students diagnosed with dyslexia at an elementary reading level, and three heuristic evaluators.

¹We use the terms *students with dyslexia* and *dyslexic learners*, following prior work [7, 16], with respect and no intent to offend.

²https://en.wikipedia.org/wiki/Sinhala_language

Co-design Participants	#	Notation	Co-design Participants	#	Notation
Dyslexia Specialist Teachers	5	T1–T5	Sinhala Language Specialists	2	SL1–SL2
Clinical Psychologist	1	CP1	Students with Dyslexia	6	S1–S6
Heuristic Evaluators	3	HE1–HE3			

Table 1: Table enumerates the co-design participant types, their number, as well as their abbreviations.

The first phase of the co-design process focused on identifying challenges and potential solutions for the lack of dyslexia-supportive applications in the Sinhala language. This was achieved through a comprehensive literature review and interviews with dyslexia specialist teachers and a clinical psychologist. The second phase of the research focused on designing and evaluating a Sinhala-specific style guide and the final phase involved designing and refining the user interfaces, which underwent continuous evaluations by heuristic evaluators.

3.1 Perspective from Teachers

To understand the problem domain, we conducted in-depth interviews with a licensed clinical psychologist with over 18 years of experience working at a special needs school and five dyslexia specialist teachers from three prominent schools in Sri Lanka. Each interview lasted approximately an hour, providing valuable insights into the challenges faced by Sinhala-speaking students with dyslexia. These interviews revealed significant gaps in awareness, training, and resources for addressing dyslexia in the country. The clinical psychologist explained that most primary school teachers are not adequately trained to support dyslexic learners, with only special education teachers receiving relevant training [21]. One teacher (T1) noted, *“Teachers in Sri Lanka don’t have standardized dyslexia materials for Sinhala. Most create their own resources to teach students”*. They also highlighted challenges in accessing international dyslexia resources due to language barriers and high subscription fees. The interviews also revealed that many students struggle with motivation to read because they find traditional learning materials challenging. The teacher (T1) added, *“Often, we only show one sentence at a time to prevent them from feeling overwhelmed and losing interest.”* Similarly, another teacher (T2) noted, *“Parents often mention that their children don’t like to read at home. They find most books difficult to read, and books with lower reading levels don’t grab their attention.”* Despite these challenges, teachers observed that incorporating small games and interactive tasks in the classroom engaged students more effectively. One teacher (T3) remarked, *“Students these days prefer mobile or computer-based learning over traditional textbooks.”* These insights show the need for a Sinhala specific interactive solution to better support students with dyslexia.

3.2 Designing a Sinhala Specific Style Guide

During this research phase, we identified the need for a dyslexia-friendly style guide tailored to the Sinhala language. The style guide was developed through a review of existing literature and consultations with a clinical psychologist, five dyslexia specialist teachers, two Sinhala language specialists, and six students as enumerated in Table 1, addressing key elements critical to readability for dyslexic learners: font type, font size, font spacing, and color [32].

3.2.1 Font Type: Sinhala is a visually complex language with distinct vowel, consonant, and special sign forms [22]. When selecting a font for Sinhala-speaking students with dyslexia, it’s crucial to consider the language’s visual characteristics. Readability depends on text presentation, making font selection vital for these learners [34]. In this study, we compared two commonly used fonts in Sri Lanka: FMAbhaya, with tighter spacing and distinct shapes, and Noto Sans Sinhala UI, with better spacing and larger characters. Research on dyslexia emphasizes the importance of letter spacing for readability, with minimal impact from letter shape [22]. To understand the students’ preferences, the six students were asked to identify the more legible and preferred font. Noto Sans Sinhala UI was determined to be the most suitable font, as it was preferred by five students. Refer to Figure 2 for a comparison of the two font types.

3.2.2 Font Size and Line Spacing: Font clarity is influenced by factors such as spacing, size, and contrast, which directly affect readability [22]. The British Dyslexia Association [4] and the Dyslexia Association of Ireland [23] recommend font sizes between 12–14 points. Based on these recommendations and feedback from students and teachers, we design the style guide with specific font sizes: 16, 14 and 12 points, for headings, subheadings and body text, respectively. To further improve readability, we follow the recommendation of maintaining line spacing at 150% of the font size [4].

3.2.3 Color: Color is a crucial element in text comprehension for individuals with dyslexia [28]. Following guidelines from the British Dyslexia Association, the style guide uses single-color backgrounds and avoids complex patterns [4]. To enhance contrast and legibility, dark-colored text on a light-colored background is recommended [37]. In the initial phase of our research, various text and background color combinations were tested with students, guided by the teachers, to determine the most effective options. Based on literature and participant feedback, the style guide primarily uses blue, yellow, and white backgrounds paired with black text for optimal readability [31].

3.3 Activities and Iterative User Interface Design:

In the third phase of our research, we identified three activities to integrate into LEXILEARN. These activities were designed to address the fundamentals of the Sinhala language, tailored for dyslexic learners with elementary reading skills. The activities underwent multiple iterations, incorporating design prototypes and feedback from participants (Table 1). The selection and integration of gamification elements were guided by dyslexia specialist teachers and a clinical psychologist. Each activity in LEXILEARN incorporates a reward system where users earn coins for completing tasks, which can then be used to unlock new avatars that act as companions

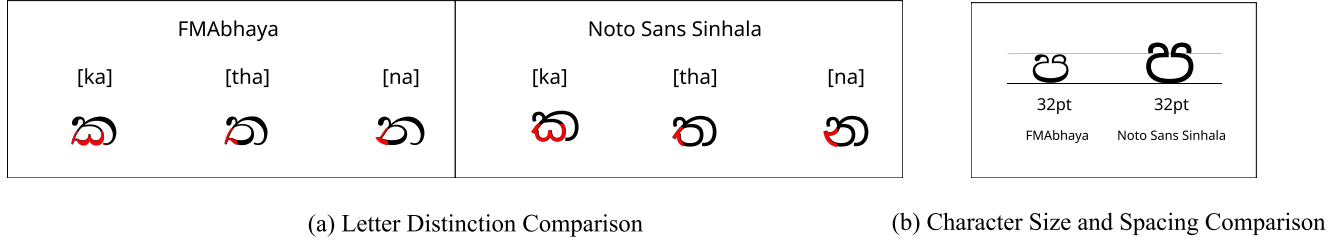


Figure 2: FMAbhaya and Noto Sans Sinhala font letter distinction comparison

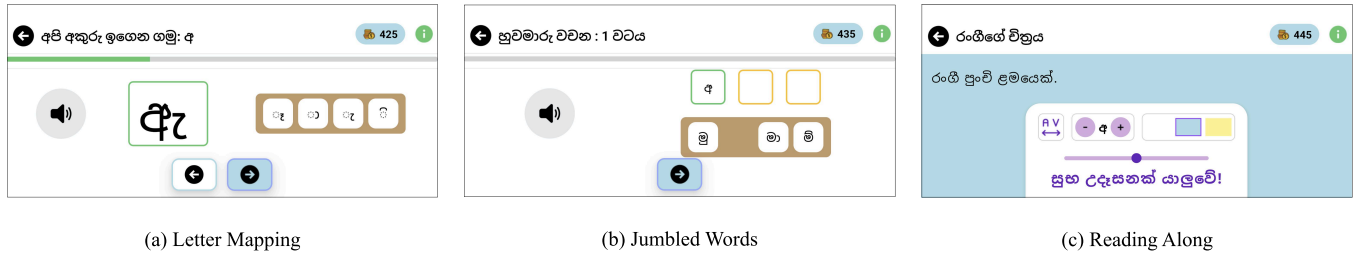


Figure 3: User Interfaces of the three activities in LexiLearn

throughout the activities. The activities are divided into levels, offering users the flexibility to skip or complete them at their own pace. This feature, emphasized by teachers, promotes autonomy by encouraging self-learning and self-monitoring, an approach supported by research [41], [15]. Positive reinforcement was central to the design, with audio and visual feedback provided after each task. Immediate feedback on incorrect attempts offers constructive guidance to facilitate learning and improvement [20], [13]. Daily streaks served as an additional incentive, motivating students to engage with the application consistently. Finally, clear auditory instructions are available throughout each activity.

3.3.1 Letter-Mapping: The first activity strengthens fundamental reading skills and phonological awareness by presenting Sinhala consonants and allowing users to add vowels to form complete sounds. Users can listen to each consonant’s pronunciation and select the correct vowel. Figure 3 (a) shows the final interface for the “Letter Mapping” activity, which aids basic reading, improves phonological awareness, and boosts short-term memory.

3.3.2 Jumbled Words: The “Jumbled Words” activity helps improve phonemic awareness and word recognition skills. Users are presented with disassembled, jumbled words and are tasked with rearranging the letters to form correct words, reinforcing letter-sound relationships. Audio pronunciations are available to support auditory recognition and phonetic understanding. Figure 3 (b) shows the finalized user interfaces for the “Jumbled Words” activity.

3.3.3 Reading Along: The “Reading Along” activity offers dyslexic learners an interactive reading experience with five books, progressing from easy to challenging, based on the high-low reading approach [40]. Users can customize text settings, such as font size, color, and spacing, to enhance readability. Figure 3 (c) shows the interface, where only one sentence is displayed at a time to minimize distractions. This design choice, informed by teachers, promotes

focus and sustained attention. The activity also includes comprehension assessments through multiple-choice questions (MCQs) to encourage interactive learning and self-assessment.

4 Evaluation

We conducted a preliminary user study to evaluate how students with dyslexia would engage with LEXILEARN as well as to understand the effect of specific gamification elements of LEXILEARN. We prioritized ethical considerations by obtaining informed consent from participants and their parents or legal guardians. Additionally, we ensured privacy and confidentiality by anonymizing all data for analysis and reporting. This study was conducted in full compliance with the ethical guidelines set by the University of Colombo School of Computing Institutional Review Board (UCSCIRB).

4.1 Participants & Procedure

The study involved eight students with dyslexia (3 male, 5 female, aged 11–15), including six from the co-design process (S1–S6) and two new participants (S7, S8). Five teachers who worked closely with the students and nine parents (one student had both parents involved) also contributed. Participants were selected by the teachers based on specific criteria: a confirmed dyslexia diagnosis, elementary reading proficiency, and informed consent.

We began the evaluation with introductory sessions for each student, their parent, and the respective teacher to introduce LEXILEARN. The app was installed on students’ devices, with a tutorial and user guide provided. Over the next seven days, students were encouraged to explore the app independently, using it at their discretion. Teachers assisted with observations during school use, while parents provided feedback based on home usage, and all parties, including researchers, were available to offer support as needed throughout the study. At the end of the study, feedback was collected through a combination of system usage logs and semi-structured interviews with students, teachers, and parents. System

logs provided quantitative insights into engagement patterns, while interviews offered qualitative perspectives.

4.2 Findings

4.2.1 Analysis of System Usage Data. The system usage data, summarized in Figure 4, highlights key insights into student engagement during the seven-day evaluation period of LEXILEARN. The data reflects usage patterns, rewards earned, and progress in unlocking avatars, as well as time spent on activities.

The system usage data showed varied engagement levels among students, ranging from 1 to 7 days of participation as shown in Figure 4(a). S3 demonstrated the highest engagement, actively using the app for all 7 days, while S7 used the app for only 1 day. Most students engaged with the app for 3–5 days, reflecting moderate overall usage. These patterns appear linked to students' proficiency levels (See Appendix A.1). Higher-level proficiency students, such as S1, S3, and S4, showed more consistent engagement, typically using the app for at least 5 days. In contrast, lower-proficiency students, like S2 and S7, engaged with the system for only 2 and 1 day, respectively. An exception to this trend was S8, who, despite having higher-level proficiency, used the app for only 3 days but engaged with all three activities during this period. These findings highlight the impact of proficiency on engagement patterns, emphasizing the need for activities tailored to diverse learner needs. During the interviews, educators affirmed this sentiment but also highlighted that other factors, such as personal interest or external support, may also play a significant role in student engagement.

4.2.2 Analysis of Gamification Elements. We analyzed the perspectives of all three groups to assess the implementation and effectiveness of the selected gamification elements in engaging students.

Table 4(b) summarizes the rewards earned by the students in the form of coins and badges, as well as their progress in unlocking avatars. Among all game elements, *coins and badges* emerged as the most motivating features for both students and teachers. In particular, students valued the tangible rewards and sense of progress, with one student stating, *"I liked earning coins; it made me feel like I was progressing through the levels fast"* (S1). Teachers echoed this sentiment, with one teacher remarking, *"Students liked the graphics and receiving coins a lot. I saw them checking the coin amount to unlock an avatar"* (T1). Badges were useful for tracking progress, as one teacher noted, *"Badges almost acted like an indicator of the student's level for me"* (T3). These elements effectively engaged students and provided positive reinforcement throughout the learning process. System usage data further supports these observations, showing a clear link between consistent participation and higher rewards, with frequent users earning more coins and badges (see Figure 4(a) and Table 4(b)).

Avatars added a layer of personalization and enjoyment to the learning experience. Students enjoyed unlocking and customizing avatars, with one stating, *"I really liked the different avatars; maybe having more to choose from would be nice"* (S3). Teachers appreciated avatars' ability to engage students, describing them as companions. *"Having the avatar as a companion through the app was a nice element. It was like a peer guiding them through the material"* (T4).

However, system usage data revealed varied adoption of this feature (Table 4(b)). While several students actively unlocked and used avatars, three participants (S2, S6, and S7) did not engage with this element. This suggests that while avatars add personalization, their appeal and utilization may need to be enhanced.

The motivational impact of these gamification elements varied with proficiency levels (see Appendix A.1). Students with strong letter recognition and vowel mapping skills, such as S1, S5, and S8, earned the maximum 3 badges in Jumbled Words and actively used coins (e.g., S3 and S8 unlocked multiple avatars). In contrast, students with basic proficiency, such as S2 and S7, earned fewer badges and showed less interest in gamified rewards like avatars. This suggests that while gamification is effective, it needs to be tailored to maintain engagement across skill levels.

Daily streaks received mixed responses. Some students found them motivating for consistent use. One parent shared, *"My child was excited to use the app every day, especially to maintain the daily streak"* (P3). However, not all students enjoyed the feature. As one student said, *"I didn't enjoy the streak much; it felt like a chore, something I had to do"* (S5). Teachers saw the potential benefits but were also concerned about the pressure it might create. One teacher noted, *"Students with learning challenges don't like to read, so if the daily streak is used as a progress measure, it can be stressful for them"* (T1).

Levels were seen as a moderate motivator, providing a sense of progress for some students but proving less engaging for others. One student commented, *"I liked having the levels; it made me feel like I was improving my Sinhala, but it didn't really make me want to use the app"* (S2). Another shared their experience with the varying difficulty: *"In some activities, going through the levels was easy, but in Reading Along, it was hard, so I just skipped some of them"* (S8). Teachers observed similar patterns, saying levels sparked initial interest but could lose appeal if progress was slow or tasks became too difficult. Therefore, the option to skip levels was appreciated for providing autonomy.

Feedback was generally appreciated for its role in providing guidance and helping students track their progress, though its repetitive delivery was sometimes disruptive. One student remarked, *"I liked it at first, but then it kept saying it a lot, and I couldn't skip it. It was a little annoying."* (S1). Teachers similarly noted the value of feedback but recommended optimizing its frequency and format. As one teacher (T1) remarked, *"Some students may not have the patience to listen to all of these audio feedbacks. I think it would be better if they had an option to mute or skip them as they want."* These insights highlight the need to improve how this gamification feature is implemented in LEXILEARN.

4.2.3 Analysis of Activities. The three activities showed varying levels of engagement and effectiveness based on system usage data (depicted in Figure 4(c)) and participant feedback. Letter Mapping was the most popular, with 23 completed levels. Its interactive design effectively supported phonics and decoding skills, engaging dyslexic learners. Jumbled Words followed with 12 completed levels, while Reading Along had the lowest engagement, with only three students attempting it. Teachers attributed this to the inherent challenges students with dyslexia encounter in decoding text. Parents also noted that additional support was often required for Reading

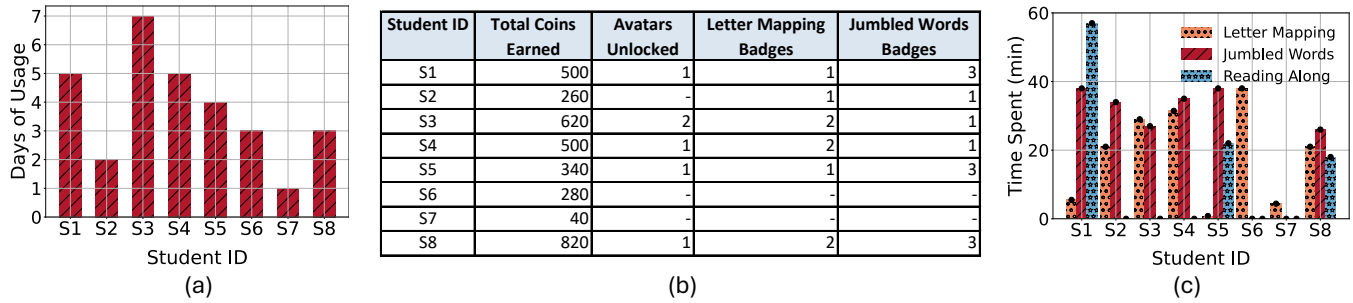


Figure 4: Figure (a) shows total days of system usage by students, (b) summarizes achievements – coins earned, avatars unlocked, and badges obtained, and (c) depicts the time spent on each of the three activities.

Along, unlike the other two activities, which students could complete independently. One parent shared, “*She needed a lot of help with it. The other two activities she completed on her own, but for Reading Along, I was helping her*” (P8). Interestingly, despite its lower usage, teachers identified Reading Along as the most impactful activity for enhancing reading comprehension. They noted its potential to foster long-term improvements in reading skills and motivation compared to traditional books. One teacher commented, “*Students are more enthusiastic to use the app for reading than the books*” (T4). While Letter Mapping and Jumbled Words were praised for building foundational skills like phonics and vocabulary, Reading Along was recognized for its ability to deepen comprehension and engage students.

Activity-specific engagement patterns showed trends based on proficiency levels (see Appendix A.1). Higher-proficiency students, like S3, S4, S5, and S8, actively participated in multiple activities, earning more coins and unlocking avatars. In contrast, lower-proficiency students, such as S6 and S7, focused on simpler activities like Letter Mapping and avoided harder ones like Reading Along. Notably, S1, a higher-proficiency student, spent the most time on Reading Along (57 minutes), showing how proficiency influenced activity choices. These findings highlight the need to balance accessibility and challenge in activities to meet diverse learner needs and encourage meaningful engagement.

The findings demonstrate that gamification elements such as coins, badges, and avatars effectively motivated and engaged students, while features like levels and feedback require adjustments to better meet diverse learning needs. The activities in LEXILEARN had varying levels of engagement and impact, with Reading Along standing out for its potential to improve reading comprehension despite lower participation. The impact of students’ proficiency levels underscores the need for designs that balance accessibility with challenge. Overall, the results indicate that gamification can effectively support learning for Sinhala-speaking students with dyslexia, offering valuable insights for creating inclusive and localized educational tools.

5 Conclusion and Future Work

We propose LEXILEARN, a game-based approach to aid learning of a visually complex language, Sinhala, for students with dyslexia. By integrating gamification elements such as coins, badges, and avatars, LEXILEARN creates an interactive and engaging learning

environment to enhance student motivation and support skill development. Our preliminary findings suggest that the gamification elements in LEXILEARN can enhance engagement, supporting existing literature.

As part of future work, we plan to conduct longitudinal studies to understand long-term effects of LEXILEARN application usage for students with dyslexia. In addition, we aim to extend the analysis to a larger group of students to closely examine any language-specific or culture-specific preferences of students that can aid to their effective learning. Overall, through this line of research, we hope to contribute to global efforts to create equitable and effective learning environments through technology for diverse learners worldwide.

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A Appendix

A.1 Student Proficiency Levels

Table 2 provides an overview of the students' abilities as assessed by educators prior to the evaluation. The table highlights the students' proficiency levels in letter recognition, vowel mapping, word recognition, and overall reading ability, categorized as "Proficient," "Basic," or "Below Basic". These ratings offer valuable insights into the strengths and areas requiring improvement for each student.

A "Proficient" rating indicates a high level of competence, where students can complete tasks with ease and without assistance. A "Basic" rating suggests that the student possesses foundational skills but may face challenges with more complex tasks. In contrast, a "Below Basic" rating reflects a need for significant support, as the student's abilities in that area fall below the expected standard.

Student ID	Age	Letter Recognition Ability	Vowel Mapping Ability	Word Recognition Ability	Reading Ability
Student 1	15	Proficient	Proficient	Proficient	Basic
Student 2	11	Basic	Basic	Basic	Below Basic
Student 3	13	Proficient	Proficient	Basic	Basic
Student 4	12	Proficient	Basic	Basic	Basic
Student 5	14	Proficient	Proficient	Proficient	Basic
Student 6	12	Basic	Basic	Basic	Below Basic
Student 7	13	Basic	Basic	Basic	Below Basic
Student 8	13	Proficient	Basic	Proficient	Basic

Table 2: Participant Overview